

How Additional Payroll Spending Impacts Fan Engagement

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Abstract

This paper studies the importance of payroll investment on fan attendance for MLB teams from 2015 to 2025, excluding 2020. To do this the paper analyzes MLB payroll data from Spotrac, a website which tracks MLB team spending, and attendance data from Baseball Reference, a baseball database. We run a multivariable regression on these and control for factors such as team success, measured in wins and playoff appearances, ballpark capacity, and metro area population. The paper found a statistically significant increase in attendance with an increase in payroll spending.

Introduction

Going into 2020 the Los Angeles Angels signed third-baseman Anthony Rendon to a 7-year, \$245 million contract. Rendon was coming off of a season in 2019 in which he led MLB in runs batted in and finished third in the most valuable player voting. After signing with the Angles, Rendon's career took a sharp downturn, struggling to stay on the field and producing significantly less effectively even when he played. Despite this, many fans showed up in Rendon jerseys, excited to watch the former superstar play. This demonstrates what I believe to be an underappreciated market in baseball, the value of a bigtime signing in the eyes of the fanbase, even if the player doesn't live up to the hype on the field.

Rendon is an extreme example but similar effects are seen with far more marginal players. In 2024, Mitch Garver signed a two year, \$24 million deal with the Seattle Mariners, a team notorious for holding relatively low payrolls and for consistently

coming up short of the playoffs. While Garver has been average at best, the signing gave the fans hope that their team was investing significantly in a good hitter.

To explore that relationship, this paper will test the correlation between team spending and attendance, with the hypothesis that fan attendance is positively correlated with team payroll. To show that the connection is deeper than the fact that spending money on a team will make that team better, we will control for team success in the regression. This will be measured in wins and postseason appearances.

The dependent variable for this paper is team attendance, which was sourced from Baseball Reference, a database of baseball statistics, past and present. Our primary independent variable is team payrolls, which came from Spotrac, a website that tracks the spending of professional sports teams. There is data going back many years but we chose to isolate 2015-2025. This was done to ensure our results more accurately portray the state of the game today, as the culture around baseball in America is constantly changing. We also chose to exclude 2020 from the data as there were no fans allowed in ballparks in 2020 due to COVID-19 restrictions.

In the regression there are a number of controls. One key control is team wins, which is a number, 0 to 162, for how many games the team won in the regular season. Another way we are controlling for success is a binary variable for whether or not the team made the postseason. Other controls include metro area population, ballpark capacity, and average age of the roster, because older rosters are generally more expensive due to players not hitting free agency until after their 6th year in the league.

We find when running our regressions that there is a significant correlation between increased payroll and higher attendance. With results significant at the 99%

level of significance, the effects are found to be positive. This demonstrates that teams will gain attendance if they increase their payroll, with the magnitude of those effects being about 8500 fans for every million dollar increase in payroll over a season.

To test for heterogeneity we checked to see if the relationship was significantly stronger in small or big markets and on winning or losing teams. In big markets, the relationship between payroll and attendance was smaller, likely due to the fact that they can sell seats more easily regardless due to larger populations. For good teams the effect of additional payroll was also less but not statistically significantly so.

The variety of factors which contribute to fan engagement in MLB teams makes it impossible to have a perfectly accurate model. While we isolated many of the factors we expected to be key in predicting attendance, the highest r-squared from any of the regressions run was 58%. This leaves 42% of the variance unaccounted for, likely made up of a number of factors including: promotions, weather, likability of players, and fan loyalty, among many others. Follow up studies exploring relationships including those factors would be fascinating to explore.

One issue with the findings of this paper is the possibility of reverse causality. It is certainly possible that teams which get high attendance are enabled to spend more on payroll. However this is mostly accounted for in our regression through the inclusion of market size as a variable, which controls for some of a teams ease in getting fans. Additionally teams more or less make all their signings before the season starts, with only a few minor free agents signing after games start, this would mean that the effects of increased attendance would only show up the following year, which is measured as a different data point in our database.

We considered running the regression with team fixed effects, but in the end teams face so much variability year to year that we decided that would not accurately portray the effects of spending properly. It is more likely using team effects that our results would show the effects of factors such as fan loyalty to a certain team rather than measuring payroll effects.

When looking into the question of whether or not increased spending increases fan engagement, it would've been interesting to observe the impacts on television viewership, merchandise purchasing, and primarily overall revenue. Unfortunately because MLB teams are privately owned, there is extremely limited data available in regards to those areas. Because attendance data is publically available, we chose to isolate that as our measure of fan engagement for this paper. It would make for a fascinating follow-up study to gather more data on other measures of engagement and observe how those are affected by payroll increases.

The remainder of this paper is organized as follows. The background section will discuss MLB payroll and roster rules, as well as established strategies for team building. The data section describes the data sources, sample construction, and key variables used in the analysis. The empirical strategies section will talk about building the regression and key omitted variables. Finally, the results section will go over the results of the regression and discuss what can be learned from this research.

Conceptual Framework

Major League Baseball (MLB), is the highest level professional baseball league in the world. It is composed of 30 teams spread out across 26 US cities and one team in Toronto, Ontario, Canada. The league will have been in existence for 150 years as of next year and has been hugely popular throughout that span. One thing that makes the business of baseball unusual in comparison to other major American sports is the lack of a salary cap or salary floor. Teams are free to spend however much they like on the players in their organization.

MLB organizations are made up of one big club, and a number of minor league teams. While minor league teams and players are invaluable to a big league club, they serve almost no purpose in increasing big-league attendance, so we will exclude minor league payroll from this paper. It would be an interesting topic for a follow up study to observe the level of impact on minor league attendance from major league investment.

With minor league players excluded we are left measuring the salary from the 26 players on a teams active roster(28 in September due to expanded rosters), or 25 players before 2021(up to 40 in September). In addition to the active roster our team payrolls include any spending on players currently on the injured list(referred to as the disabled list prior to the 2019 season). This is to capture the value of hype and hope on a team's fans as injured players can still give fans reason to believe in their team in the long run and get excited about going to games

Due to this lack of restriction, there is a large amount of variance in team payrolls. In 2025 the lowest payroll team is the Miami Marlins, who are spending \$67,906,371 on their 26-man active roster and injured players who were previously on

the 26-man roster. On the high end, the Los Angeles Dodgers are spending \$332,246,524, 389% higher than the Marlins. This separation creates a massive chasm between the best and worst teams in the league in terms of talent on the field.

The advantages of spending money on your payroll are generally beneficial to the quality of your play on the field. Getting more good players will make your team better, so assuming teams are reasonably able to evaluate talent, higher payrolls will lead to better teams the majority of the time.

Data

The team's total payroll expenditure on big league players(`total_payroll`) is the independent variable in our regression. Payroll comes from Spotrac, a website which tracks salaries in professional sports leagues. The mean of payroll is 143 million dollars, with a standard deviation of 56 million, meaning payroll also varies highly between organizations.

Home game ballpark attendance(`attendance`) is our dependent variable. Attendance data was drawn from Baseball Reference, who has a database including MLB attendance as well as other external factors for each MLB organization. The mean of attendance is 2,256,206 people over 81 games, which works out to 27,854 fans per game over 81 home games. This represents 65% of the mean stadium capacity, while leagues like the NFL and NBA see crowds filling 90% or more regularly. There is also a standard deviation of 732,304 fans which is fairly high variance, meaning attendance varies widely from team to team and year to year.

The average age variable(age) represents the average of all players on the rosters age, weighted by playing time. This is calculated and displayed on Spotrac with their payroll data. We chose to include age in the regression because older teams will generally be more expensive, thus being correlated with payroll, so including it would violate our assumptions for unbiasedness.

Team wins(wins) was drawn from the Baseball Reference as well, and simply measures how many of the team's 162 games the team won. We chose to not include losses in the regression because it would cause multicollinearity with wins, as would winning percentage, since teams play the same number of games in every season in the sample.

Whether or not a team made the postseason(postseason) is a valuable control in this regression. This obviously would have ramifications on fan support as making the postseason and having the chance to play for a championship is the ultimate goal of most sports fans. Because this paper is only measuring attendance in the regular season, we chose to only measure postseason success in terms of making or missing the playoffs, placing no value on how far the team makes it, since fans are unaware of that during the regular season and thus wouldn't impact their decision to attend games. Additionally, we chose to include the wild card game and wild card series as regular playoff games, since they occur outside of the regular season schedule and fit into the playoff structure. This also allows for a more consistent definition of a playoff team. One downside to this is that the playoff structure changed following the 2021 season to allow for one additional wild card team per league. However, we have chosen to not account for this in this paper because the goal is to measure how fans respond and fans don't

care how many teams are in the playoffs as long as their team is one of them. To measure playoff appearances we made a binary variable that is equal to 1 if the team played in a playoff or wild card game that season, and 0 if they failed to appear in any sort of playoff game.

The amount of fans a team's home stadium can hold(capacity) was drawn from a January 13th article by Shane Jackson on the BetMGM website. In addition to the 27 active ballparks that have been in use for their current teams throughout our sample, we added in the capacity of Globe Life Field, the Texas Rangers' home through the 2019 season, Tropicana Field, the Rays' home through 2024, and The Oakland Coliseum, the Athletics' home through 2024. Neutral site games were not included in the sample.

For the metro area population variable(metropop2020), values are pulled from the US census data from 2020. The only exception is Toronto, which being a Canadian city isn't included in the US census. Toronto's data came from a number of sources which agreed on the same number. Additionally 2020 was chosen for a multitude of reasons, the primary being that it's the most recent US census, and thus will include the most reliable data. Additionally it's in the middle of our sample so should be fairly accurate across years. We chose to use just the single year's population for simplicity purposes, and because all other years are estimates based on data from census years. Another factor of choosing populations was identifying metropolitan areas as a team's market, rather than just the city they call home. This decision was made to more accurately portray the size of a fanbase, as suburban fans will still regularly attend games in their nearby city. Obviously, it won't perfectly capture the fanbase size as fans exist outside the team's primary market, but it's as close to an estimate as we can

capture. This method will likely underestimate market size for historic franchises such as the Cardinals and Yankees, while overestimating market size for teams which are new to their markets, like the Nationals, or share a metro area with another team, such as the A's. One decision made to mitigate that issue was to assign the Angels to just being from Anaheim and its surrounding area, rather than the entire Los Angeles metro. We chose to include the entire bay area for the A's and Giants, although considerations were given to splitting it.

Empirical Strategy

To test the question of if higher payroll leads to higher attendance we need to regress attendance against payroll. This will show the impact increasing payroll by one dollar has on fan attendance for MLB teams. Our initial regression equation is equation one.

$$attendance = \beta_0 + \beta_1 payroll \quad (1)$$

In order to isolate payroll as the cause and avoid omitted variable bias, we also need to control for the quality of the team and advantages they face to get fans in the ballpark. To do this, we are adding average age, regular season wins, whether or not the team made the postseason, stadium capacity, and the population of the teams metro area to our regression. Adding said variables gets us equation two.

$$attendance = \beta_0 + \beta_1 payroll + \beta_2 avg_age + \beta_3 wins + \beta_4 postseason + \beta_5 capacity + \beta_6 metropop$$

(2)

Attendance and payroll are both very large numbers, and both have fairly high variance so to control for that we also want to run the same regression but taking the natural log of both attendance and payroll. This will result in a linear relationship. Another advantage of using natural log is it shows how a one percent increase in payroll affects the percent change in attendance, far more valuable when working with such large numbers. When doing so we get equation three.

$$\ln_attendance = \beta_0 + \beta_1 \ln_payroll + \beta_2 avg_age + \beta_3 wins + \beta_4 postseason + \beta_5 capacity + \beta_6 metropop$$

(3)

Due to concerns that wins may be a flawed measure, because fans care very little about the difference between 65 and 70 wins over the course of a 162 game season, we also created a variable for whether or not a team was over a .500 winning percentage(goodteam). If a team is over .500 they have a reasonable shot at the playoffs, and below almost no chance, so the new metric should capture fairly well the amount fans believe their team can make the playoffs. Goodteam is a better measure of competitiveness than postseason, because it rewards being in contention, and teams in close playoff chases usually get fans to show up. We then multiply goodteam by payroll to observe the effects of adding payroll on only better than .500 teams.. Adding in goodteam and replacing wins and postseason yield the following regression equation, equation 4.

$$\ln_attendance = \beta_0 + \beta_1 \ln_payroll + \beta_2 avg_age + \beta_3 goodteam + \beta_4 capacity + \beta_5 metropop$$

(4)

Another area where we may see effects from different situations is teams with different market sizes. We created a binary variable to measure the size of the market as big or small(bigmarket), where small markets were above the median population of 4,570,686 people. Then multiplying that with payroll we were able to observe the effects of being in a big or small market, see equation five.

$$attendance = \beta_0 + \beta_1 payroll + \beta_2 avg_age + \beta_3 wins + \beta_4 postseason + \beta_5 capacity + \beta_6 bigmarket + \beta_7 bigpayroll$$

(5)

Results

When we run the regression on equation one we find the results pictured in figure one:

Source	SS	df	MS	Number of obs	=	270
Model	6.5874e+13	1	6.5874e+13	F(1, 268)	=	225.23
Residual	7.8383e+13	268	2.9247e+11	Prob > F	=	0.0000
				R-squared	=	0.4566
				Adj R-squared	=	0.4546
Total	1.4426e+14	269	5.3627e+11	Root MSE	=	5.4e+05

attendance	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
total_payroll	.0087558	.0005834	15.01	0.000	.0076072	.0099045
_cons	1002697	89775.36	11.17	0.000	825942	1179451

(1)

This shows a positive correlation between payroll and attendance that is statistically significant at the 99% level of significance. The β_0 coefficient states that if a team spent \$0 on payroll they would expect 1,002,697 fans. This is impossible as MLB has minimum contracts, but sets a baseline to see the effects of payroll. The β_1 coefficient states that for each one dollar increase in payroll, a team would gain 0.0088 fans over the course of the season. When scaled up to larger contracts, this represents an increase of 8,755 fans for an increase of one million dollars in payroll.

When we run equation two, we get the results shown below in figure two.

Source	SS	df	MS	Number of obs	=	270
Model	8.2869e+13	6	1.3811e+13	F(6, 263)	=	59.17
Residual	6.1388e+13	263	2.3341e+11	Prob > F	=	0.0000
				R-squared	=	0.5745
				Adj R-squared	=	0.5647
Total	1.4426e+14	269	5.3627e+11	Root MSE	=	4.8e+05

attendance	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
total_payroll	.0082944	.0007407	11.20	0.000	.006836	.0097528
avg_age	-102630	41246.08	-2.49	0.013	-183844.6	-21415.48
wins	16449.1	3730.291	4.41	0.000	9104.065	23794.14
capacity	40.06612	7.408495	5.41	0.000	25.47861	54.65363
postseason	26634.11	94497.63	0.28	0.778	-159434.1	212702.3
metropop2020	-.0183416	.007715	-2.38	0.018	-.0335326	-.0031506
_cons	983149.4	1091595	0.90	0.369	-1166229	3132528

(2)

This regression adds controls to isolate the impact of payroll on attendance. Adding the controls has a very minor effect on the results of β_1 . B_1 is still statistically significant at the 95% level and the impact of a payroll increase of one million dollars has an impact of 8294 fans over the course of the season. With all the controls included

the r-squared value is still only 0.5745, meaning this regression explains 57.45% of the variance in attendance, leaving almost 43% of the variance unexplained.

When we run equation three the results are captured in figure three.

Source	SS	df	MS	Number of obs	=	270
Model	21.2612487	6	3.54354145	F(6, 263)	=	62.49
Residual	14.9136967	263	.056706071	Prob > F	=	0.0000
				R-squared	=	0.5877
				Adj R-squared	=	0.5783
Total	36.1749454	269	.134479351	Root MSE	=	.23813

ln_attendance	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ln_payroll	.6617532	.0487925	13.56	0.000	.5656795	.7578269
avg_age	-.0756724	.0202474	-3.74	0.000	-.11554	-.0358047
wins	.0080369	.0018398	4.37	0.000	.0044144	.0116595
capacity	.0000152	3.64e-06	4.18	0.000	8.05e-06	.0000224
postseason	.0068237	.0465938	0.15	0.884	-.0849206	.098568
metropop2020	-9.76e-09	3.67e-09	-2.66	0.008	-1.70e-08	-2.54e-09
_cons	3.055439	.7806256	3.91	0.000	1.518368	4.592511

(3)

Regression 3 finds that a one percent increase in payroll leads to a 0.66% increase in fan attendance. These findings are relevant at the 99% level.

One key variable we were unable to control for was ticket prices. There simply isn't enough reliable data and are too many other factors involved in ticketing to include in our regression. This is important though, because improving the team would lead to an increase in ticket prices, as well as quantity sold, due to increased demand for tickets. Additionally the increase in ticket prices would counteract some of the increase in ticket sales, due to higher prices driving would-be customers away.

The next regression includes the goodteam variable, rather than wins and losses, to try to capture whether the team being competitive affects these results tremendously, with results shown in figure four.

Source	SS	df	MS	Number of obs	=	270
Model	7.8448e+13	6	1.3075e+13	F(6, 263)	=	52.25
Residual	6.5809e+13	263	2.5022e+11	Prob > F	=	0.0000
				R-squared	=	0.5438
				Adj R-squared	=	0.5334
Total	1.4426e+14	269	5.3627e+11	Root MSE	=	5.0e+05

attendance	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
total_payroll	.0088736	.001028	8.63	0.000	.0068495	.0108977
avg_age	-57189.42	41553.72	-1.38	0.170	-139009.7	24630.89
goodteam	410790.2	183692.3	2.24	0.026	49095.43	772485
capacity	40.7785	7.710923	5.29	0.000	25.5955	55.9615
metropop2020	-.0161241	.0080799	-2.00	0.047	-.0320336	-.0002147
goodpayroll	-.0008193	.0012119	-0.68	0.500	-.0032056	.0015669
_cons	793597	1150634	0.69	0.491	-1472031	3059225

(4)

The effects of being a good or bad team are fairly minimal on the difference payroll makes in attendance. The difference is less than 800 fans for every \$1 million spent and is not statistically significant. This was slightly surprising to me but can be explained by the fact that fans want to see their teams be better regardless of the level at which that improvement happens. It serves to confirm my hypothesis, that regardless of success, teams benefit from increased investment by growing fan interest.

When we control for big or small markets the regression yields the results shown below in figure five.

Source	SS	df	MS	Number of obs	=	270
Model	8.5355e+13	7	1.2194e+13	F(7, 262)	=	54.24
Residual	5.8901e+13	262	2.2481e+11	Prob > F	=	0.0000
				R-squared	=	0.5917
				Adj R-squared	=	0.5808
Total	1.4426e+14	269	5.3627e+11	Root MSE	=	4.7e+05

attendance	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
total_payroll	.0103617	.001097	9.45	0.000	.0082016	.0125219
avg_age	-103282.6	40784.7	-2.53	0.012	-183590.1	-22975.12
wins	17023.06	3685.466	4.62	0.000	9766.16	24279.96
capacity	36.74037	7.194093	5.11	0.000	22.57477	50.90597
postseason	51867.94	93005.71	0.56	0.578	-131265.8	235001.7
bigmarket	244590.4	177096	1.38	0.168	-104122.2	593302.9
bigpayroll	-.0032625	.0012291	-2.65	0.008	-.0056828	-.0008423
_cons	829567.1	1071465	0.77	0.439	-1280211	2939345

(5)

This shows that the effects of increasing payroll are larger in small markets than in big markets. The difference is about 3000 fans for every million dollars spent, and is a statistically significant difference at the 95% level. These differences are likely related to the fact that fans are easier to get to go to games when there are more people in the area.

Overall, these results show that increasing payroll does increase attendance, regardless of success or market size, by a significant amount. However if each additional ticket sold because of these changes was sold for \$100, a slightly generous estimate, the team would make back about \$850,000 on their \$1,000,000 investment. This is an underestimation however as the team could also use the increased demand to increase the price of tickets on all tickets sold. Additionally this only captures the increased attendance, and doesn't measure the value it has for television or radio

viewership, or other forms of profit like merchandise sales. There are clearly valuable returns to increasing spending as a MLB organization although the scale of that value is not clearly determined from this paper.

Conclusion

Many owners in baseball choose not to spend less money on their team if they don't see winning a championship as a possibility, but the results of this paper prove that there are benefits to increasing investment regardless. There is certainly room for follow up research to focus on how payroll affects other measures of fan engagement. We were pleased with the results of this paper and our hypothesis was confirmed, spending more money on an MLB team leads to higher attendance at home games that season.